

Change of Variables in Probability

Math Foundations – Concept 28

1 Setup

Let X be a continuous random variable taking values in an open set $U \subseteq \mathbb{R}^n$ with probability density function (PDF) $f_X : U \rightarrow [0, \infty)$. Let $g : U \rightarrow V$ be a transformation; we wish to find the density of $Y = g(X)$.

Definition 1 (Diffeomorphism). *A map $g : U \rightarrow V$ between open subsets of $\mathbb{R}^n$ is a C^1 diffeomorphism if g is a bijection, $g \in C^1(U)$, and $g^{-1} \in C^1(V)$. Equivalently, g is C^1 and its Jacobian matrix $J_g(x) = (\partial g_i / \partial x_j)_{ij}$ is invertible for every $x \in U$.*

Definition 2 (Push-forward density). *If X has density f_X and $Y = g(X)$ has a density, we call that density the push-forward of f_X along g and denote it $f_Y = g_{\#}f_X$.*

2 Main results

Theorem 1 (1D change of variables). *Let X be a continuous random variable with density f_X on an open interval $(a, b) \subseteq \mathbb{R}$. Let $g : (a, b) \rightarrow (c, d)$ be a C^1 function with $g'(x) \neq 0$ for all $x \in (a, b)$. Then g is strictly monotone, hence bijective, and $Y = g(X)$ has density*

$$f_Y(y) = \frac{f_X(g^{-1}(y))}{|g'(g^{-1}(y))|}, \quad y \in (c, d).$$

Proof. Because g' is continuous and never zero on the connected interval (a, b) , either $g' > 0$ everywhere or $g' < 0$ everywhere. We treat the two monotone cases via the cumulative distribution function (CDF) approach.

Case 1: g strictly increasing. Then $g : (a, b) \rightarrow (c, d)$ is a bijection with C^1 inverse $g^{-1} : (c, d) \rightarrow (a, b)$, and for any $y \in (c, d)$,

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y)).$$

Differentiating in y (chain rule; F_X is differentiable with $F'_X = f_X$, and g^{-1} is differentiable with $(g^{-1})'(y) = 1/g'(g^{-1}(y))$ by the inverse function theorem),

$$f_Y(y) = F'_Y(y) = f_X(g^{-1}(y)) (g^{-1})'(y) = \frac{f_X(g^{-1}(y))}{g'(g^{-1}(y))}.$$

Since $g' > 0$, we have $|g'| = g'$, so this equals the claimed formula.

Case 2: g strictly decreasing. Then for $y \in (c, d)$,

$$F_Y(y) = P(g(X) \leq y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y)).$$

Differentiating,

$$f_Y(y) = -f_X(g^{-1}(y)) (g^{-1})'(y) = -\frac{f_X(g^{-1}(y))}{g'(g^{-1}(y))}.$$

Since $g' < 0$, $-1/g' = 1/|g'|$, recovering the claimed formula.

In both cases, $f_Y(y) = f_X(g^{-1}(y))/|g'(g^{-1}(y))|$, completing the proof. $\square$

Theorem 2 (Multivariate change of variables). *Let X be a continuous random vector on an open set $U \subseteq \mathbb{R}^n$ with density f_X , and let $g : U \rightarrow V$ be a C^1 diffeomorphism onto an open set $V \subseteq \mathbb{R}^n$. Then $Y = g(X)$ has density*

$$f_Y(y) = f_X(g^{-1}(y)) \cdot |\det J_{g^{-1}}(y)| = \frac{f_X(x)}{|\det J_g(x)|} \Big|_{x=g^{-1}(y)}, \quad y \in V.$$

Proof sketch. For any Borel set $B \subseteq V$, using $g^{-1}(B) \subseteq U$ and the multivariate substitution theorem from analysis,

$$P(Y \in B) = P(X \in g^{-1}(B)) = \int_{g^{-1}(B)} f_X(x) dx = \int_B f_X(g^{-1}(y)) |\det J_{g^{-1}}(y)| dy.$$

Because B was arbitrary, the integrand is the density of Y . $\square$

3 Canonical example

Let $X \sim \text{Uniform}[0, 1]$ and $Y = -\log X$. Then $g(x) = -\log x$ is C^1 on $(0, 1)$ with $g'(x) = -1/x$, $|g'(x)| = 1/x$, and $g^{-1}(y) = e^{-y}$. By Theorem ??,

$$f_Y(y) = \frac{1}{1/e^{-y}} = e^{-y}, \quad y > 0,$$

i.e. $Y \sim \text{Exp}(1)$. This identity underlies inverse-CDF sampling.